

Q1 - 25 July - Shift 1

A body of mass 0.5 kg travels on straight line path with velocity $v = (3x^2 + 4)\text{m/s}$. The net workdone by the force during its displacement from $x = 0$ to $x = 2\text{ m}$ is :

- (A) 64 J (B) 60 J
(C) 120 J (D) 128 J

Space for your notes:

Q2 - 25 July - Shift 2

A bag of sand of mass 9.8 kg is suspended by a rope. A bullet of 200 g travelling with speed 10 ms^{-1} gets embedded in it, then loss of kinetic energy will be

- (A) 4.9 J (B) 9.8 J
(C) 14.7 (D) 19.6 J

Space for your notes:

Q3 - 26 July - Shift 1

As per the given figure, two blocks each of mass 250g are connected to a spring of spring constant 2Nm^{-1} . If both are given velocity v in opposite directions, then maximum elongation of the spring is :



- (A) $\frac{v}{2\sqrt{2}}$ (B) $\frac{v}{2}$
(C) $\frac{v}{4}$ (D) $\frac{v}{\sqrt{2}}$

Space for your notes:

Q4 - 26 July - Shift 2

Questions

MathonGo

A body of mass 8 kg and another of mass 2 kg are moving with equal kinetic energy. The ratio of their respective momenta will be :

- (A) 1:1 (B) 2:1 (C) 1:4 (D) 4:1

Space for your notes:

Q5 - 27 July - Shift 1

Sand is being dropped from a stationary dropper at a rate of 0.5 kgs^{-1} on a conveyor belt moving with a velocity of 5 ms^{-1} . The power needed to keep belt moving with the same velocity will be :

- (A) 1.25 W (B) 2.5 W
(C) 6.25 W (D) 12.5 W

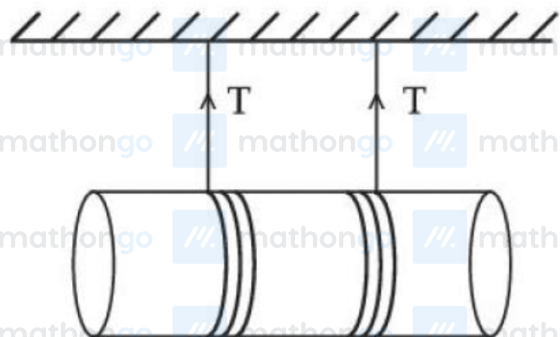
Space for your notes:

Q6 - 27 July - Shift 2

A solid cylinder length is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should by unbinding the strings to achieve a speed of 4 ms^{-1} , is.....cm.

(take $g = 10 \text{ ms}^{-2}$)

Space for your notes:



Q7 - 27 July - Shift 2

#MathBoleTohMathonGo

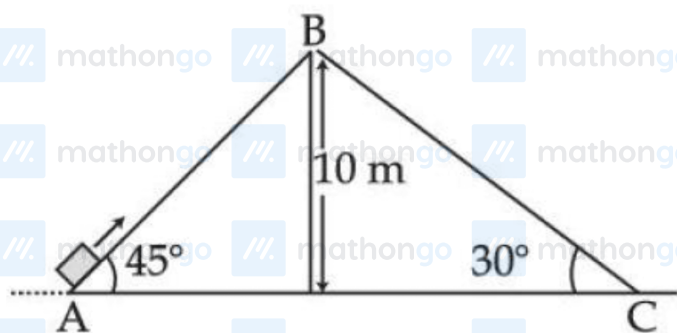
Questions

MathonGo

Two inclined planes are placed as shown in figure.

Space for your notes:

A block is projected from the Point A of inclined plane AB along its surface with a velocity just sufficient to carry it to the top Point B at a height 10 m. After reaching the Point B the block slides down on inclined plane BC. Time it takes to reach to the point C from point A is $t(\sqrt{2} + 1)$ s. The value of t is.....(use $g = 10 \text{ m/s}^2$)



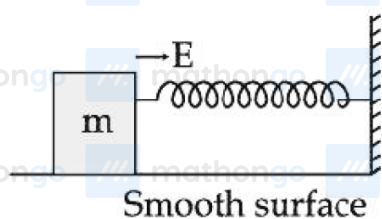
Q8 - 28 July - Shift 1

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Questions

MathonGo

A block of mass 'm' (as shown in figure) moving with kinetic energy E compresses a spring through a distance 25 cm when, its speed is halved. The value of spring constant of used spring will be $nE \text{ Nm}^{-1}$ for $n =$ _____.

*Space for your notes:***Q9 - 28 July - Shift 2**

A bullet of mass 200 g having initial kinetic energy 90 J is shot inside a long swimming pool as shown in the figure. If its kinetic energy reduces to 40 J within 1s, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



- (A) 45 m (B) 90 m
(C) 125 m (D) 25 m

*Space for your notes:***Q10 - 28 July - Shift 2**

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Sun light falls normally on a surface of area 36 cm^2 and exerts an average force of $7.2 \times 10^{-9} \text{ N}$ within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of incident light is

- (A) $25.92 \times 10^2 \text{ W/cm}^2$ (B) $8.64 \times 10^{-6} \text{ W/cm}^2$
(C) 6.0 W/cm^2 (D) 0.06 W/cm^2

Space for your notes:

Q11 - 29 July - Shift 2

If momentum of a body is increased by 20%, then its kinetic energy increases by :

- (A) 36% (B) 40%
(C) 44% (D) 48%

Space for your notes:

Answer Key

Q1 (B)

Q2 (B)

Q3 (B)

Q4 (B)

Q5 (D)

Q6 (120)

Q7 (2)

Q8 (24)

Q9 (A)

Q10 (D)

Q11 (C)

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Q1 (B)

$$v_i = 3(0^2) + 4 = 4 \quad \cong \quad x = 0$$

$$v_F = 3(2)^2 + 4 \quad \cong \quad x = 2$$
$$= 16$$

$$W = \Delta K = \frac{1}{2}m(16^2 - 4^2)$$

$$= \frac{1}{2} \times \frac{1}{2} (256 - 16)$$

$$= \frac{240}{4} = 60 \text{ J}$$

Q2 (B)

$$P_i = P_f \text{ (no any external force)}$$

$$0.2 \times 10 = 10 \times v$$

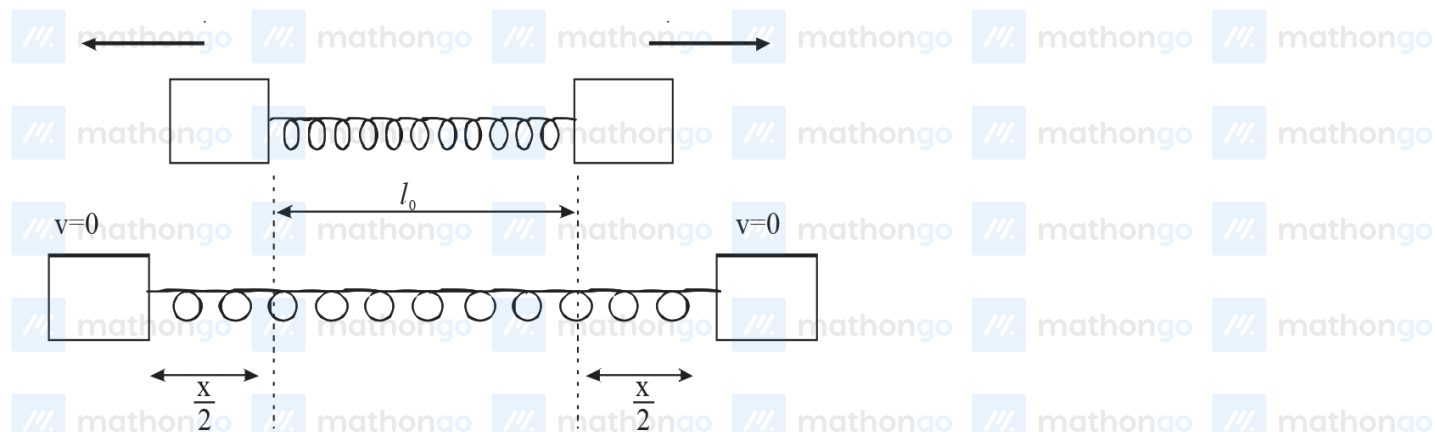
$$v = 0.2 \text{ m/sec}$$

$$\text{Loss in K.E.} = \frac{1}{2} \times (0.2) \times 10^2 - \frac{1}{2} \times 10 (0.2)^2$$

$$= \frac{1}{2} \times 10 \times (0.2) [10 - 0.2]$$

$$= 9.8 \text{ J}$$

Q3 (B)



using energy conservation

$$\frac{1}{2}mv^2 \times 2 = \frac{1}{2}kx^2$$

$$\Rightarrow \frac{1}{4}v^2 = \frac{1}{2} \times 2 \times x^2$$

$$\therefore x = \frac{v}{2}$$

Q4 (B)

$$K.E = \frac{p^2}{2m}$$

$$K_1 = \frac{P_1^2}{2(8)} ; K_2 = \frac{P_2^2}{2(2)}$$

$$K_1 = K_2$$

So,

$$4P_2^2 = P_1^2$$

$$\frac{P_1}{P_2} = 2$$

Q5 (D)

$$\text{Thrust} = \lambda V_{\text{rel}}$$

$$= 2.5 \text{ N}$$

$$\text{Now, Power} = F \times V = 12.5 \text{ W}$$

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Q6 (120)

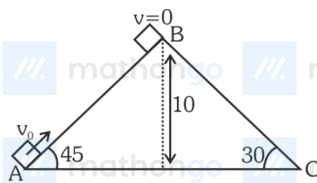
From energy conservation

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2} \frac{mR^2}{2} \omega^2$$

$$10h = \frac{16}{2} + \frac{16}{4} \Rightarrow h = 1.2\text{m} = 120\text{cm}$$

Q7 (2)



From E.C. $= \frac{1}{2}mv_0^2 = mgh$

$$v_0 = 10\sqrt{2}$$

For A $\rightarrow$ B

at B, $v = 0$

$$a = -g \sin 45^\circ = \frac{-10}{\sqrt{2}}$$

$$v = u + at_1 \Rightarrow 0 = 10\sqrt{2} - \frac{10}{\sqrt{2}}t_1 \Rightarrow t_1 = 2\text{sec}$$

For B $\rightarrow$ C

$$s = ut_2 + \frac{1}{2}at_2^2$$

$$\frac{10}{\sin 30^\circ} = \frac{1}{2}(10 \sin 30^\circ)t_2^2$$

$$t_2 = 2\sqrt{2}$$

So total time

$$T = t_1 + t_2$$

$$= 2\sqrt{2} + 2$$

$$= 2(\sqrt{2} + 1)\text{sec}$$

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Q8 (24)

Using work – energy theorem

$$W_{\text{net}} = (K_f - K_i)$$

$$\Rightarrow -\frac{1}{2}Kx^2 = \frac{1}{2}m\left(\frac{v}{2}\right)^2 - \frac{1}{2}mv^2 = \frac{E}{4} - E$$

$$\Rightarrow \frac{1}{2}Kx^2 = \frac{3E}{4} \Rightarrow K = \frac{3E}{2x^2}$$

$$\Rightarrow K = \frac{3E}{2 \times \left(\frac{1}{4}\right)^2} = 24E$$

$$n = 24$$

Q9 (A)

$$\text{Using } mv = \sqrt{2mk}$$

$$u = \frac{1}{0.2} \sqrt{2 \times 0.2 \times 90} = 30 \text{ m/s}$$

$$v = \frac{1}{0.2} \sqrt{2 \times 0.2 \times 40} = 20 \text{ m/s}$$

$$a = \frac{20 - 30}{1} = -10 \text{ m/s}^2$$

$$s = \frac{-u^2}{2a} = 45 \text{ m}$$

Q10 (D)

$$\frac{I}{C} \times \text{area} = \text{force}$$

$$\frac{I}{C} \times 36 \times 10^{-4} = 7.2 \times 10^{-9}$$

$$I = \frac{7.2 \times 10^{-9} \times 3 \times 10^8}{36 \times 10^{-9} \times 10}$$

$$= \frac{6 \times 10^{-1}}{10^{-3}}$$

$$I = 6 \times 10^2 \frac{\text{W}}{\text{m}^2}$$

$$= 0.06 \frac{\text{W}}{\text{cm}^2}$$

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Q11 (C)

$$P' = P + \frac{20}{100}P = 1.2P$$

$$\% \text{ change in KE} = \frac{K' - K}{K} \times 100$$

$$= \left(\frac{\frac{P'^2}{2m} - \frac{P^2}{2m}}{\frac{P^2}{2m}} \right) \times 100$$

$$= [(1.2)^2 - 1] \times 100$$

$$= 44\%$$